

Dental Caries

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Disease Mechanism of Dental Caries

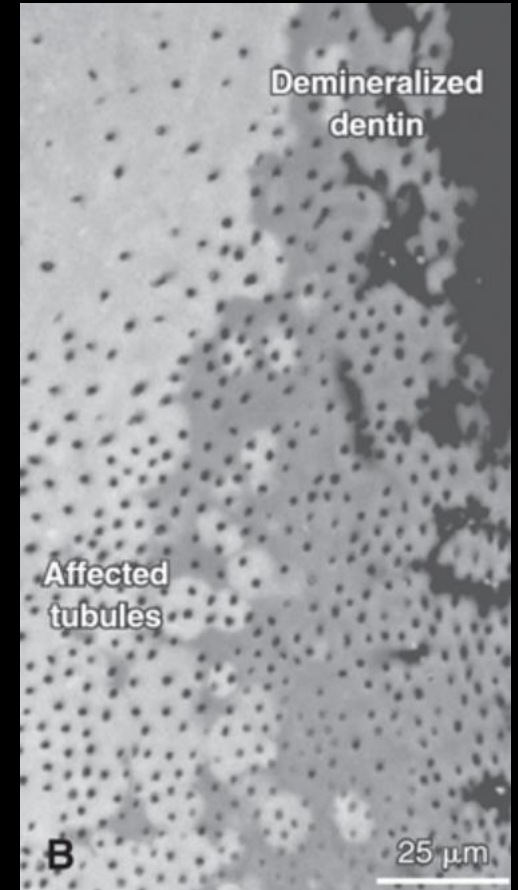
- **Multifactorial Etiology:**
 - Involves **tooth, dental biofilm, and diet**
 - Influenced by **saliva, fluoride exposure, immune system, bacterial exposure duration, and socioeconomic status**
- **Role of *Streptococcus mutans*:**
 - Ferments dietary sugars (especially fructose)
 - Produces **lactic acid** → lowers pH → **demineralization**



Light photomicrograph showing dead tracts in a radicular carious lesion that appear dark under transmitted light.

Disease Mechanism of Dental Caries

- **Demineralization Process:**
 - Begins **beneath enamel surface**
 - Leaves a thin, intact surface layer
 - Not linear – involves cycles of **demineralization and remineralization**
- **Modifying the Cycle:**
 - Reduce sugar intake
 - Improve oral hygiene
 - Use topical fluoride
- **Clinical Presentation:**
 - **Early lesion:** white, chalky
 - **Arrested lesion:** dark (brown/black)
 - **Dentin** demineralizes faster than enamel
 - Progression can lead to **pulp necrosis**



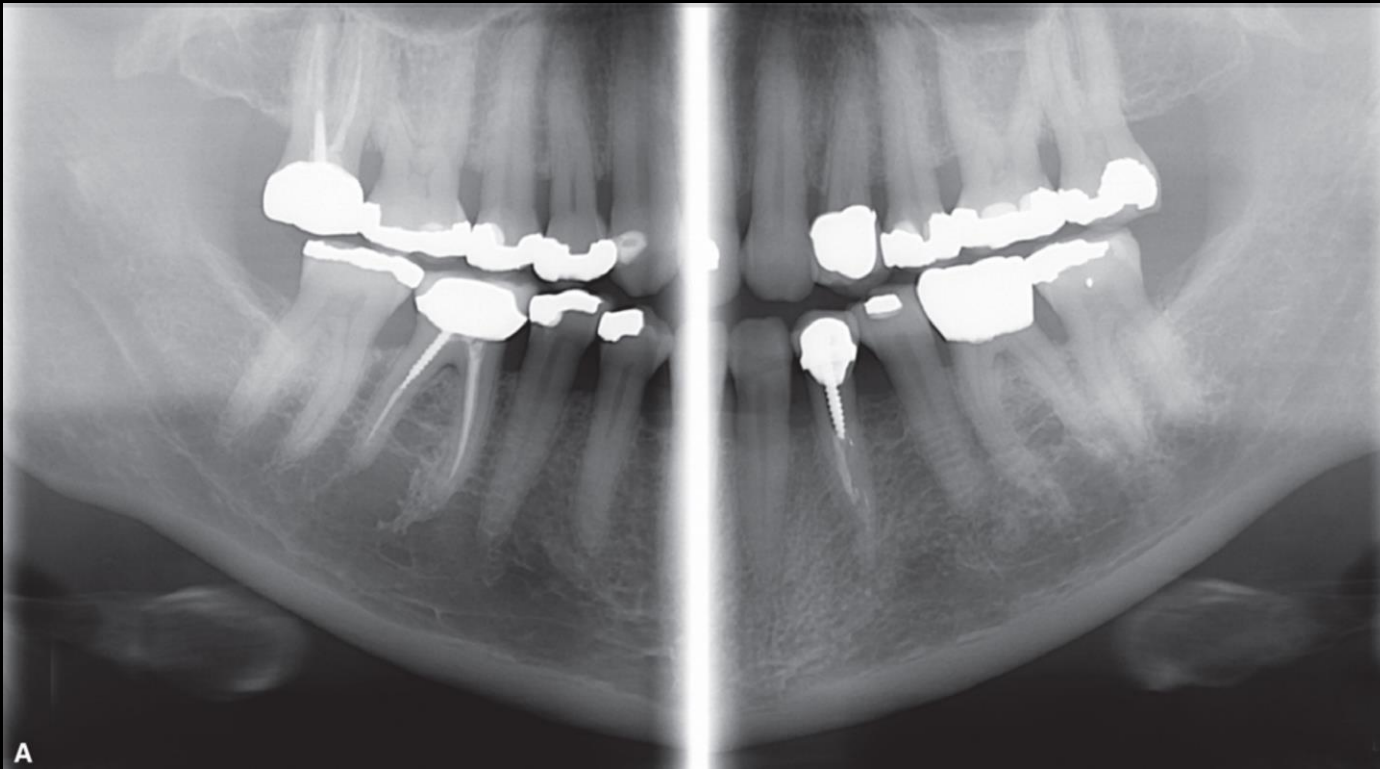
Scanning electron micrograph showing empty tubules under a carious lesion.

Role of Imaging in Caries Detection

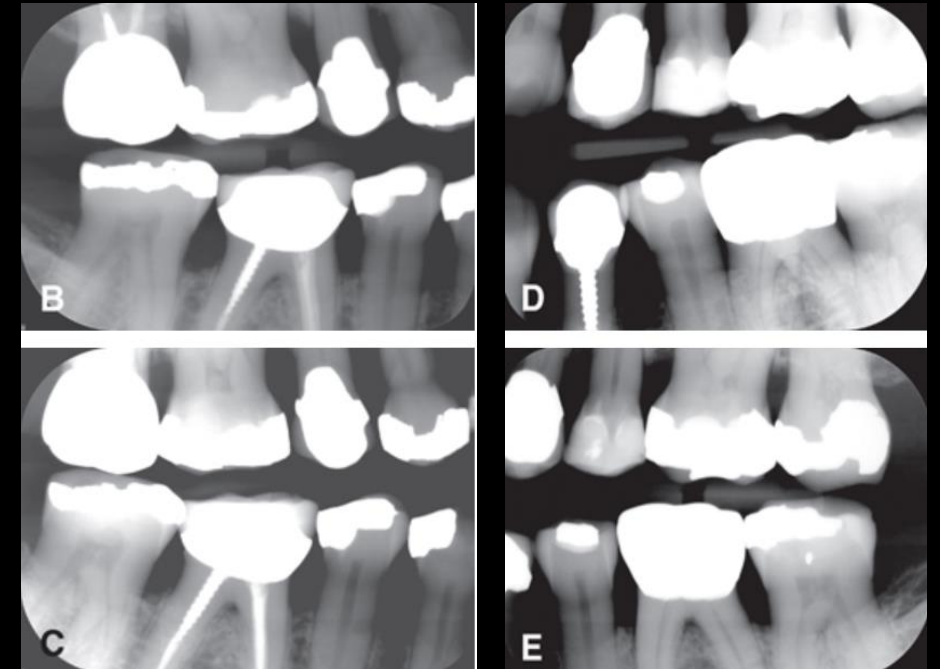
- **Clinical Limitations:**
 - Occlusal and smooth surface caries may be visible
 - **Interproximal caries** often missed unless cavitated
- **Preferred Imaging:**
 - **Intraoral bitewing radiographs:** Best for detecting interproximal caries
 - **Panoramic images:** Low resolution, not ideal for caries detection
- **Radiographic Appearance:**
 - Demineralized areas appear **radiolucent (dark)**
 - Cannot distinguish **active vs. arrested** lesions on a single image
- **Imaging Intervals (Based on Risk):**
 - **High risk:** 6–12 months (children), 6–18 months (adults)
 - **Low risk:** 12–24 months (children), 18–36 months (adolescents), 24–36 months (adults)
- **Extraoral Bitewings:**
 - Alternative for patients with gag reflex, disabilities, or anatomical constraints
 - Less reliable than intraoral bitewings

Role of Imaging in Caries Detection

Extraoral Bitewings



Intraoral Bitewings



Intraoral Imaging: Digital Sensors vs. Film

- **Types:**
 - Solid-state (CMOS)
 - Photostimulable phosphor plates (PSP)
- **Advantages:**
 - Immediate image acquisition
 - Equivalent sensitivity to film for most caries
 - PSP similar in size to film → easier placement
- **Challenges:**
 - Solid-state sensors can be bulky and uncomfortable
 - Smaller active area may require multiple images
 - Requires high-quality monitor and subdued lighting for viewing

Intraoral Imaging: Digital Sensors vs. Film

- **Intraoral Film**
- **Film Sizes:**
 - ANSI size 2 (adolescents/adults)
 - ANSI size 0 or 1 (children)
- **Viewing Requirements:**
 - Mounted with dark borders
 - Viewed on light box with magnification
 - Subdued lighting enhances detail
- **Caution:**
 - Avoid ANSI size 3 film due to distortion from bending

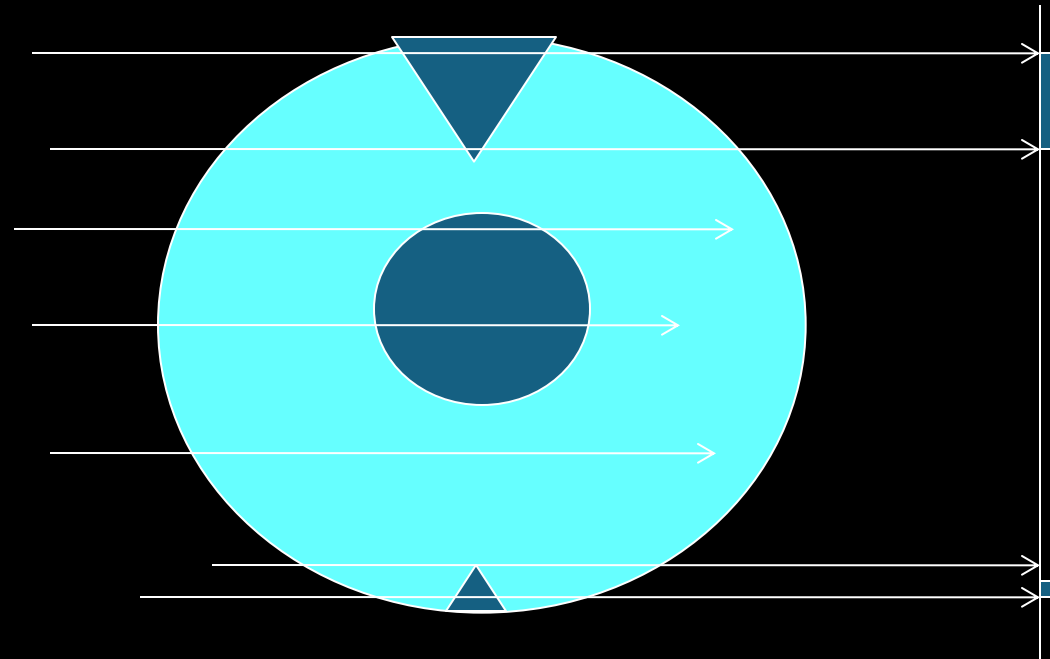
Detection of Carious Lesions

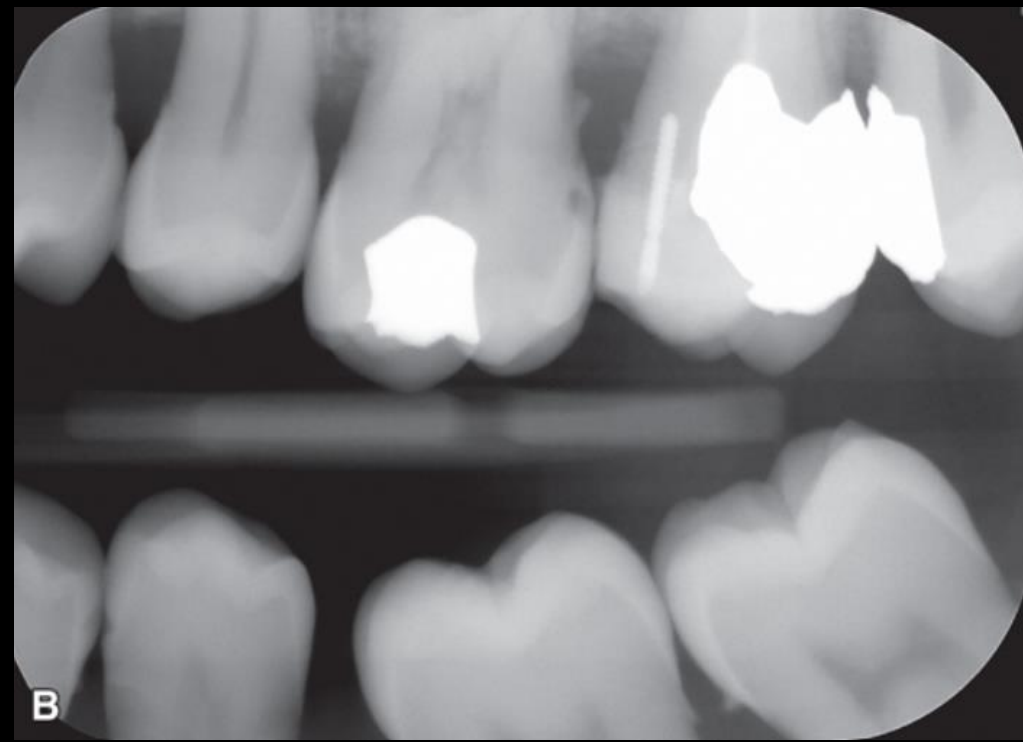
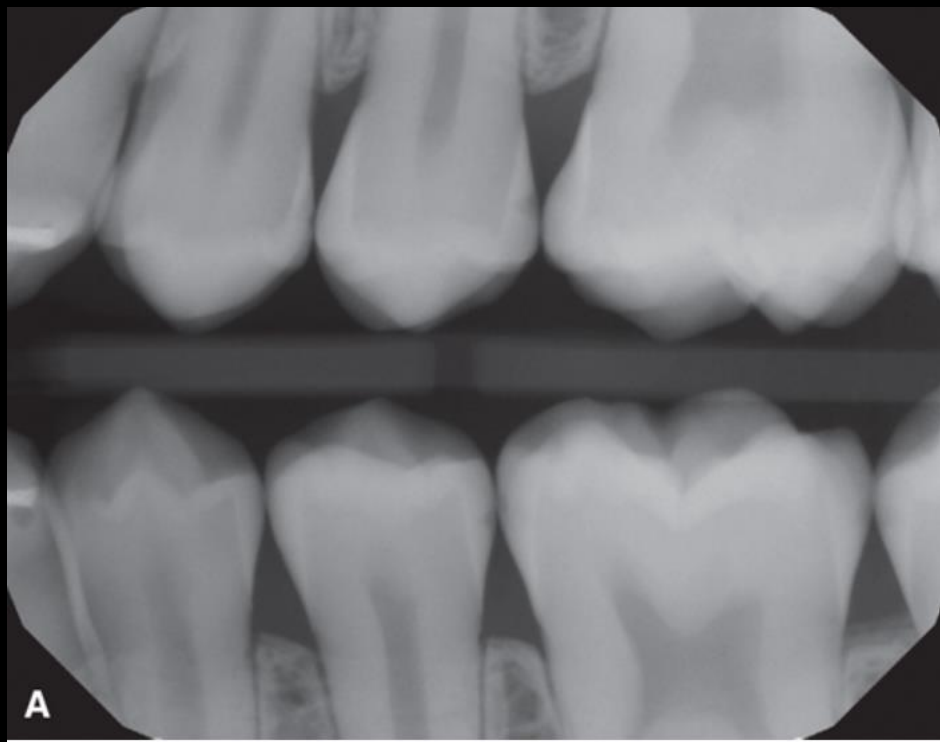
Proximal Surfaces

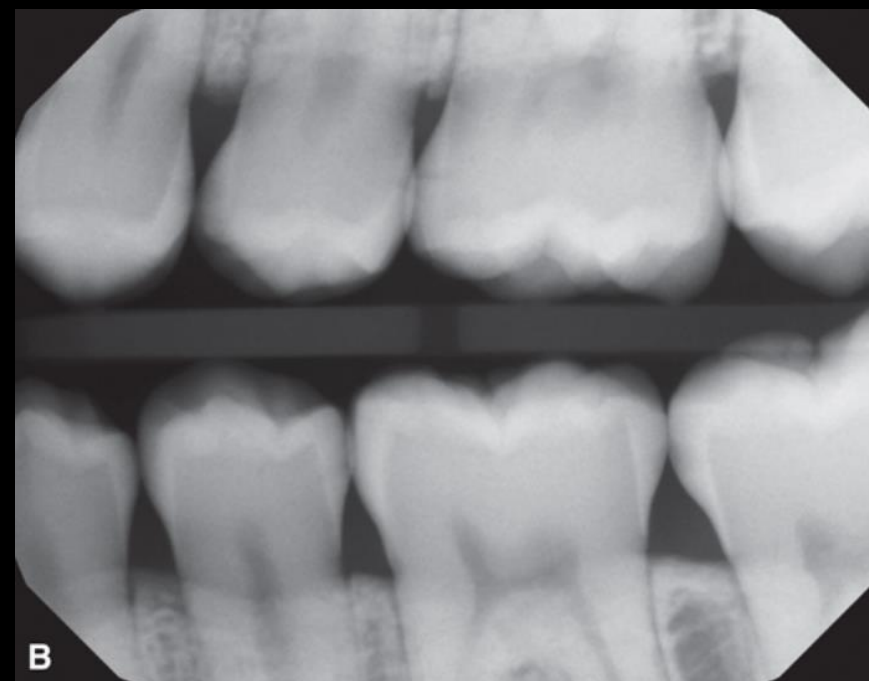
- **Enamel Caries Shape:**
 - Classic **triangular radiolucency**: base at surface, tip at DEJ
 - Other shapes: band, rectangle, semicircular notch
- **Dentin Involvement:**
 - Second triangle: base at DEJ, tip toward pulp
 - More aggressive due to lower mineral content
- **Common Location:**
 - Between contact points to near gingival margin
 - Important to distinguish from **cervical burnout**
- **Radiographic Limitations:**
 - Must lose ~35% mineral content to be visible
 - True depth often greater than what appears on image

In general, at least ~35% of the hard tissue must be affected before you see a change on a radiograph.

Caries generally extends into tooth farther than the radiograph detects.

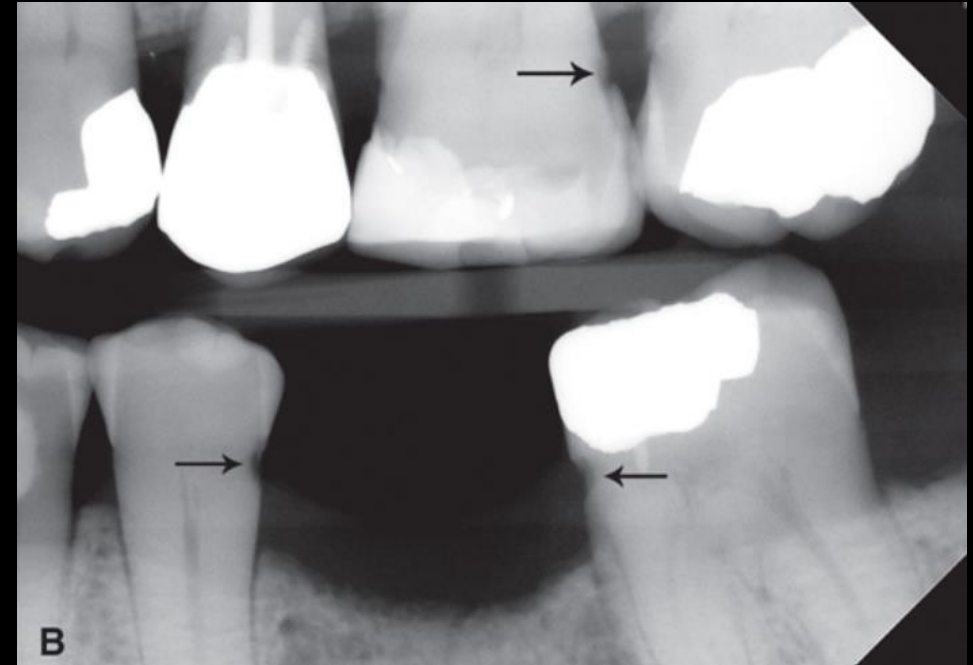
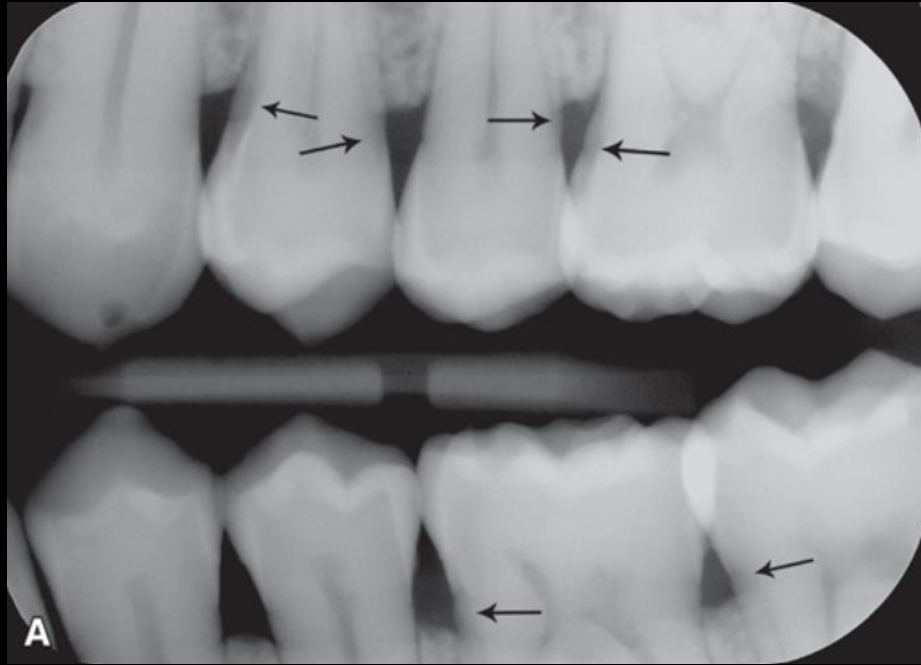






Differentiating Caries from Cervical Burnout

- **Cervical burnout:** radiolucent artifact near CEJ
- Caused by:
 - Tooth anatomy (concavities)
 - X-ray beam angulation
 - Thin enamel near CEJ
- **Common in:**
 - Single-rooted teeth (concavities)
 - Multi-rooted teeth (root overlap)
- **Avoiding misdiagnosis:**
 - Identify PDL spaces
 - Adjust horizontal angulation
 - Rely on clinical experience



False-Positive Interpretations

- **Observer variability:** <100% agreement in caries diagnosis
- **Common causes:**
 - Cervical burnout
 - Overlapping proximal surfaces
 - Inexperience
- **Prevention:**
 - Proper training
 - Correct x-ray beam angulation
 - High-quality images

Caries Staging – ICDAS/ICCMS System

- **Key Points:**
- **Merged ICDAS/ICCMS stages:**
 - **0:** Sound surface
 - **RA:** Initial stage (RA1–RA3)
 - **RB:** Moderate stage (RB4)
 - **RC:** Extensive stage (RC5–RC6)
- **Radiographic criteria:**
 - RA1: Outer enamel
 - RA2: Inner enamel ± DEJ
 - RA3: Outer third of dentin
 - RB4: Middle third of dentin
 - RC5–RC6: Inner third/pulp with cavitation

Treatment Considerations

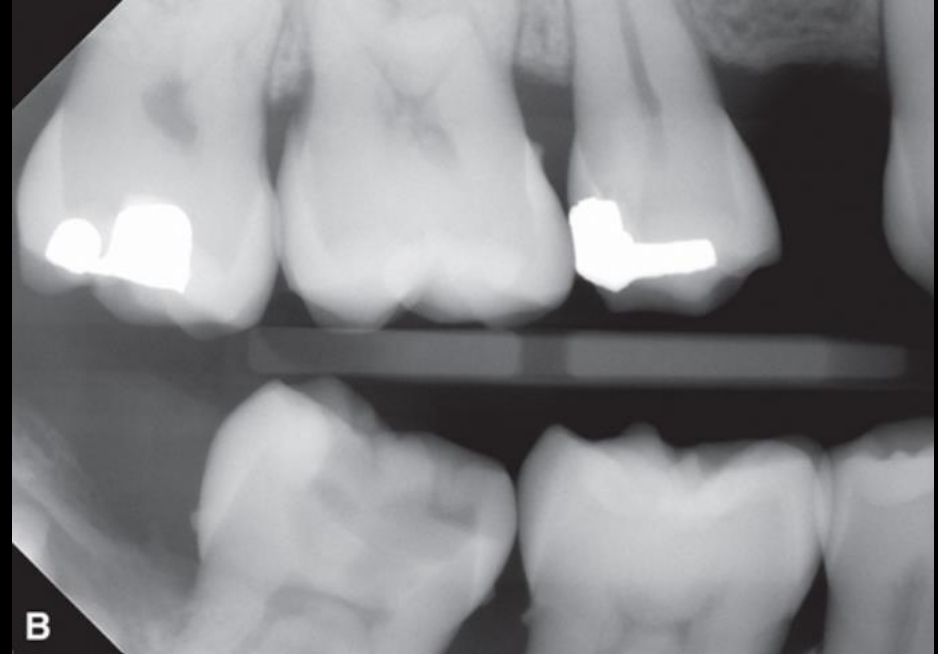
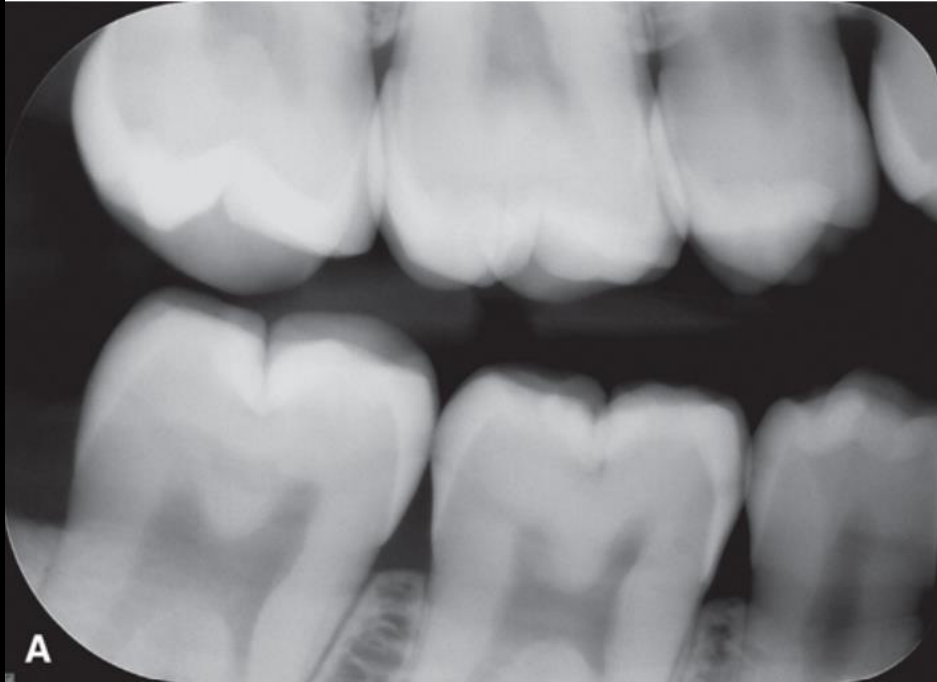
- **Treatment Depends On:**
 - Caries risk status
 - Lesion depth
 - Presence of cavitation
- **Non-Surgical Management:**
 - ICCMS **RA stage** (enamel or outer third of dentin)
 - Includes **oral hygiene instruction** and **topical fluoride**
- **Surgical Management:**
 - Required for **cavitated lesions**
 - Lesions in **middle third of dentin (RB stage)** or deeper
- **Follow-Up Strategy:**
 - Imaging schedule based on **caries risk**
 - New images should match original for accurate comparison
 - Progression = active lesion → may require surgical intervention

Occlusal Surfaces

Detection & Appearance

- Common in **children and adolescents**
- **Radiographic detection** difficult in enamel due to thickness
- Lesion becomes visible when it reaches **dentin** (semicircular shape at DEJ)
- **Clinical signs:** Chalky white (new), brown/black (older)
- **Imaging** used to assess **depth**, not early detection





Occlusal Caries – False Negatives & Mach-Band Effect

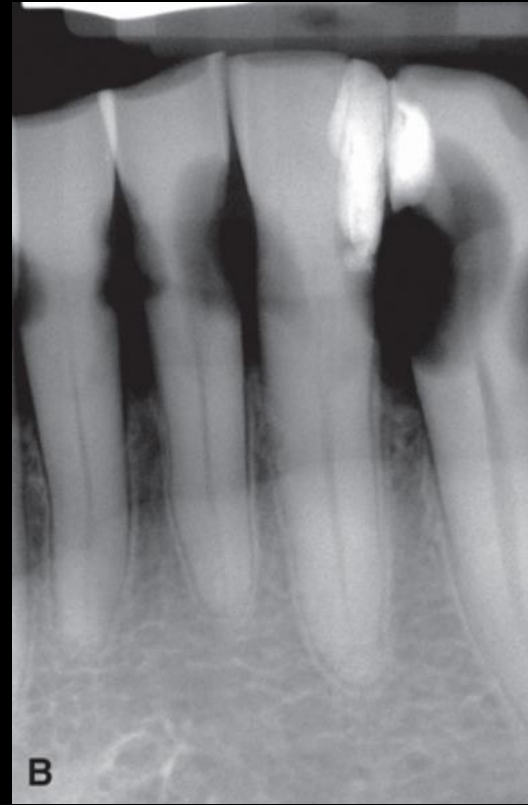
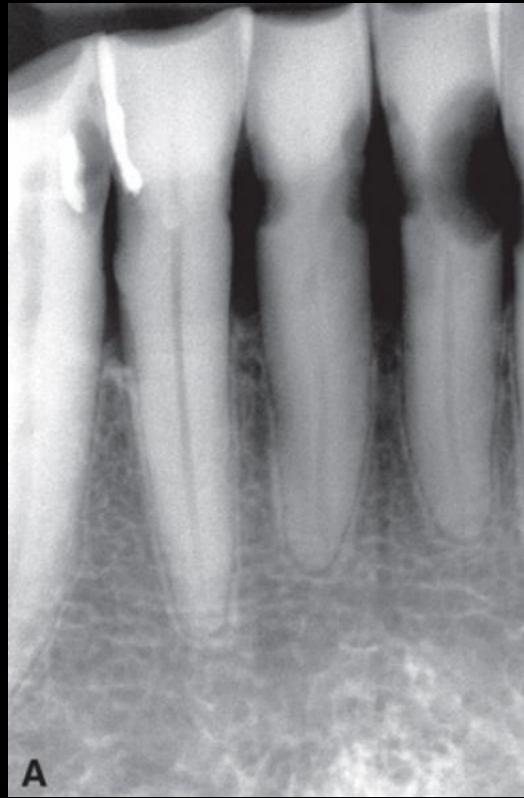
- False negatives common due to enamel superposition
- Mach-band effect: Optical illusion at DEJ
- Caused by contrast between enamel and dentin
- **Test:** Mask enamel, if band disappears, no caries; if not, caries likely
- Monitor suspicious areas without clinical signs



Buccal, Lingual & Root Surface Caries

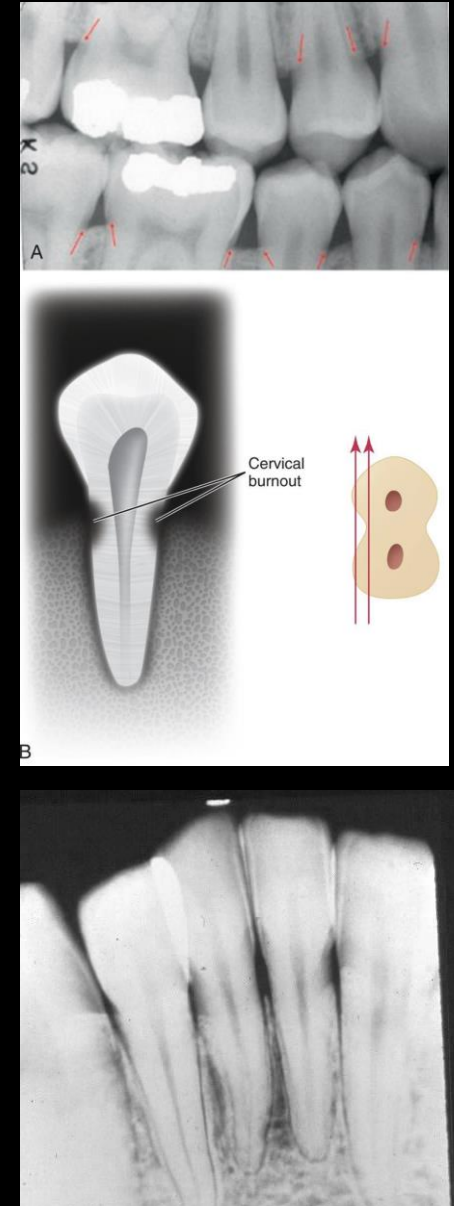
- **Buccal/Lingual:** Circular/oval, well-defined; hard to localize on single image
- Use **tube-shift technique** to determine location
- **Root caries:** Common with **gingival recession** or **bone loss**
- Cementum is thin → rapid progression
- Differentiate from **cervical burnout** via clinical exam and bone level





Cervical Burnout

- A radiolucent area seen at the cervical region of teeth
- Occurs due to:
 - Thinner root structure at the CEJ
 - Lack of bone coverage in this area
 - X-ray beam attenuation differences
- Appears as a diffuse, radiolucent band near the CEJ
- Root outline remains visible
- Disappears or changes with different horizontal angulation



Cervical Burnout

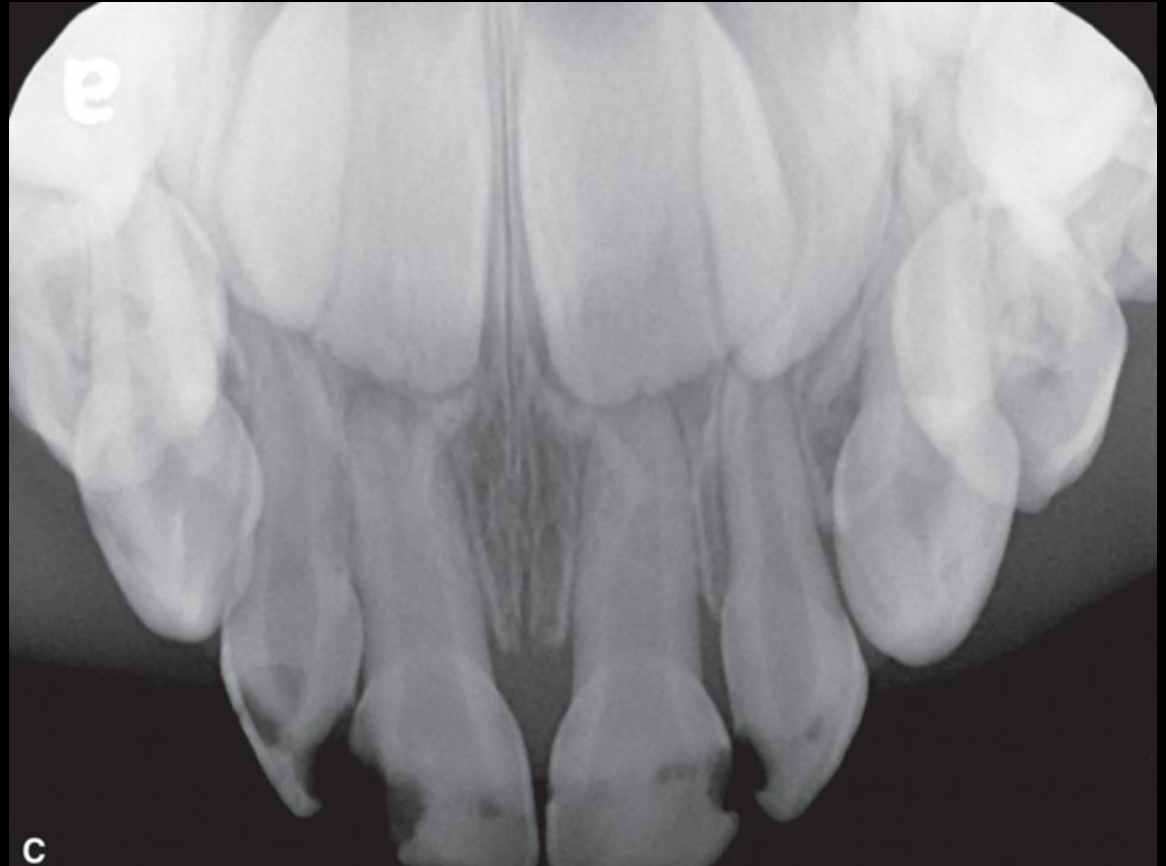
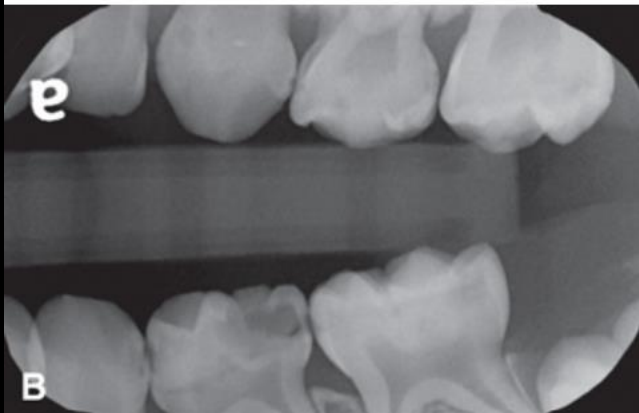
- Appears as a diffuse, radiolucent band near the CEJ
- Edges are gradual and less defined
- Root outline remains visible
- Occurs with normal bone height
- Radiographic appearance changes with horizontal angulation

Cervical Caries

- Appears as a well-defined radiolucent lesion
- Sharp, irregular borders
- Root outline may be obscured
- Often associated with bone loss or gingival recession
- Radiographic appearance remains consistent with angulation

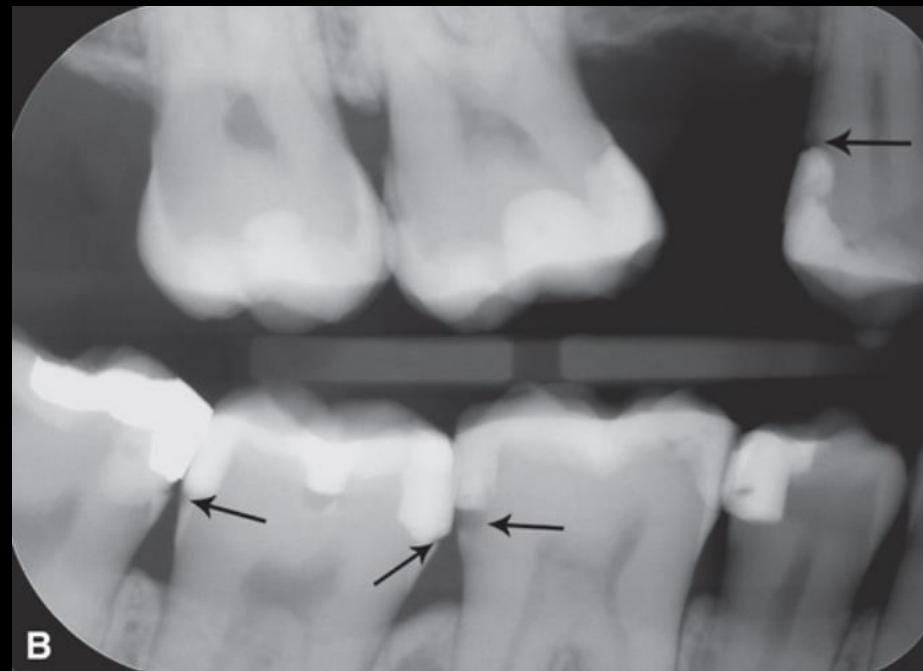
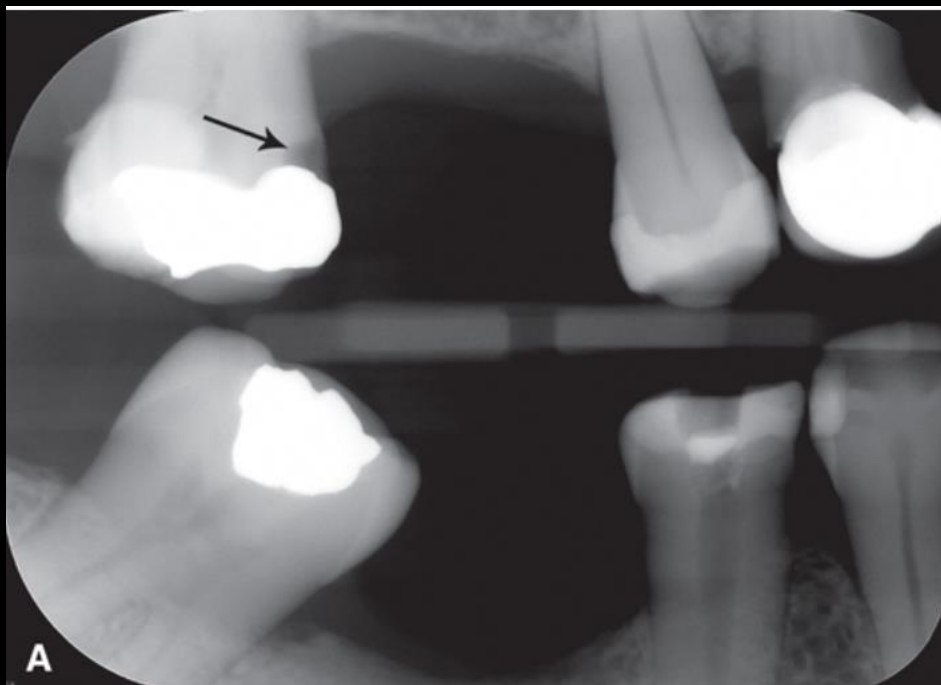
Rampant Caries – Pediatric and Adult Forms

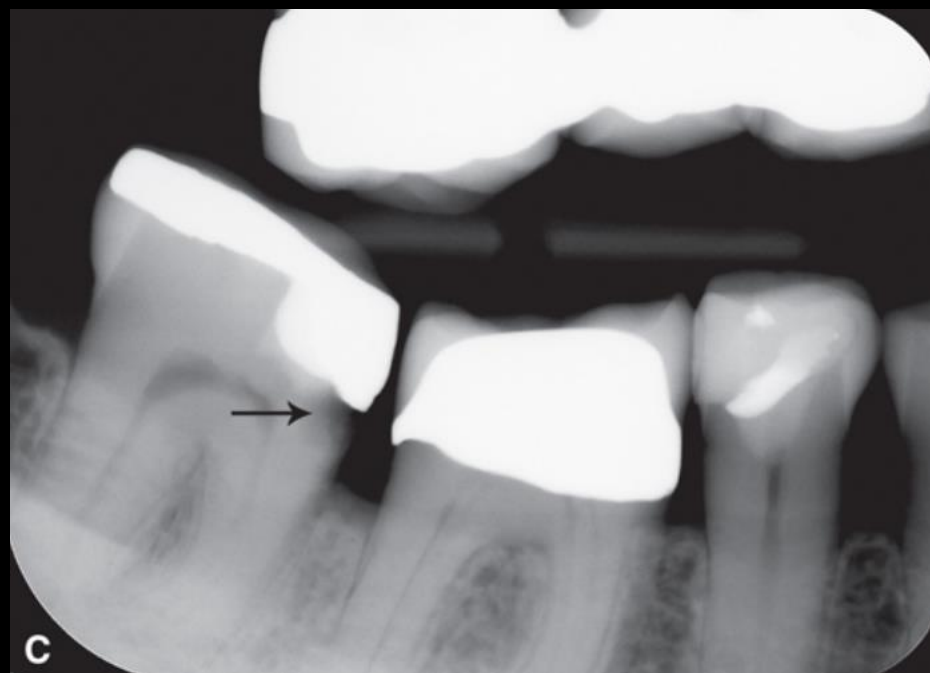
- **Rampant caries:** Rapid, widespread decay
- **Children:** Often due to poor hygiene and sugary drinks (e.g., bottle at bedtime)
- **Adults:** Common in **xerostomia** (e.g., post-radiation therapy)
- Affects **smooth surfaces** near cervical areas
- Prevention: **Fluoride**, saliva substitutes



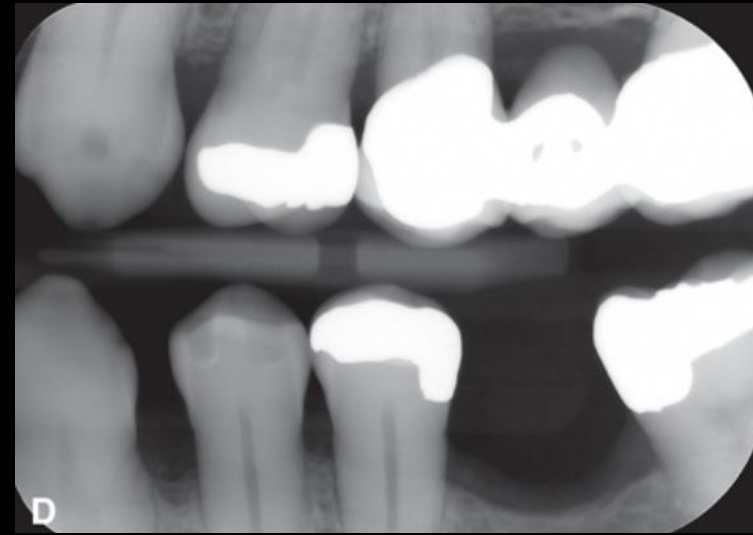
Caries Associated with Dental Restorations

- **Recurrent (Secondary) Caries:**
 - New demineralization at the **margin of a restoration**
 - Caused by **defective restorations** or **poor hygiene**
 - Appears as **radiolucency** at restoration-tooth interface
- **Residual Caries:**
 - **Unremoved carious tissue** left intentionally or unintentionally
 - Often part of **indirect pulp capping** strategy
- **Imaging Challenges:**
 - Improper angulation can **mask lesions**
 - Use **bitewing images** with correct vertical angulation
 - Consider second image with different angle
- **Material Confusion:**
 - **Radiolucent liners/bases** may mimic caries
 - **Older composites** may appear carious but have **well-defined borders**









Alternative Diagnostic Tools & Treatment Considerations

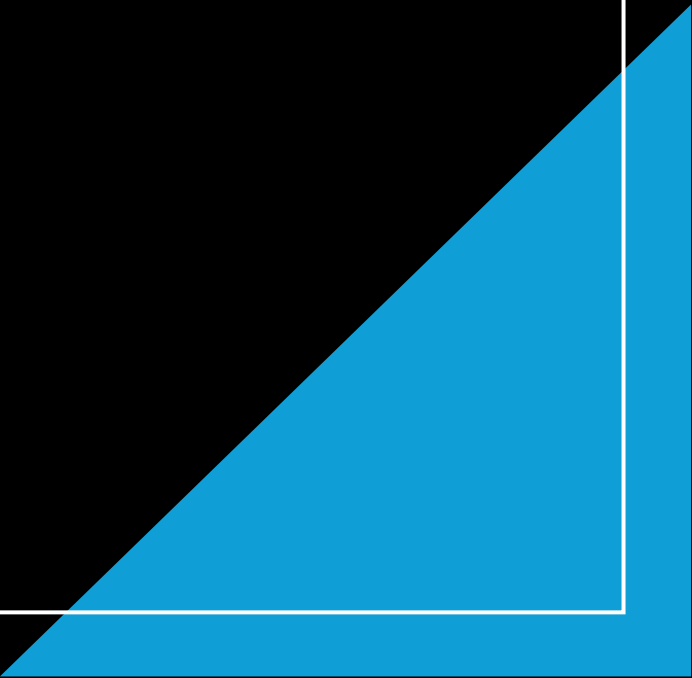
- **Alternative Diagnostic Tools:**

- **Light-induced fluorescence:** Detects demineralization on smooth surfaces
- **Lasers, transillumination, electrical conductance:** Used for occlusal/interproximal surfaces
- **AI in dentistry:**
 - Enhances diagnostic accuracy
 - Should **complement**, not replace, clinical judgment

- **Treatment Considerations:**

- Based on **oral hygiene, caries risk, lesion depth, and number of lesions**
- **Enamel-only lesions:** Non-invasive care (fluoride, hygiene, diet)
- **Early dentin lesions:** May or may not need restoration
- **Deep dentin/cavitated lesions:** Require surgical removal and restoration

Periodontal diseases

A solid blue right-angled triangle is positioned in the bottom right corner of the slide, with its hypotenuse running from the bottom left towards the top right.

The Periodontium & Periodontal Disease

Overview

- **Periodontium:** Gingiva, periodontal ligament (PDL), cementum, alveolar bone
- **Periodontal diseases:** Inflammatory conditions affecting supporting structures
- **Untreated disease** → bone loss, soft tissue damage, tooth loss
- **Prevalence:** ~42% of U.S. adults aged 30–79 have periodontitis
- **Higher risk** in older adults and lower socioeconomic groups

Disease Mechanism of Periodontal Disease

- **Initiated by dental plaque biofilm**
- Bacteria (gram-negative rods, spirochetes) → toxins + host inflammatory response
- Inflammation causes:
 - **Tissue destruction**
 - **Bone resorption**
 - **Pocket formation**
- Disease progresses in **bursts**: active vs. quiescent phases
- Influenced by: **age, stress, smoking, hormones, systemic disease**

Clinical Signs of Periodontal Disease

- **Gingivitis:** Swelling, redness, bleeding
- **Periodontitis:**
 - Pocket formation
 - Gingival recession
 - Clinical attachment loss
 - Tooth mobility
 - Bleeding on probing, pus
- Often **painless** → patients unaware until advanced stages

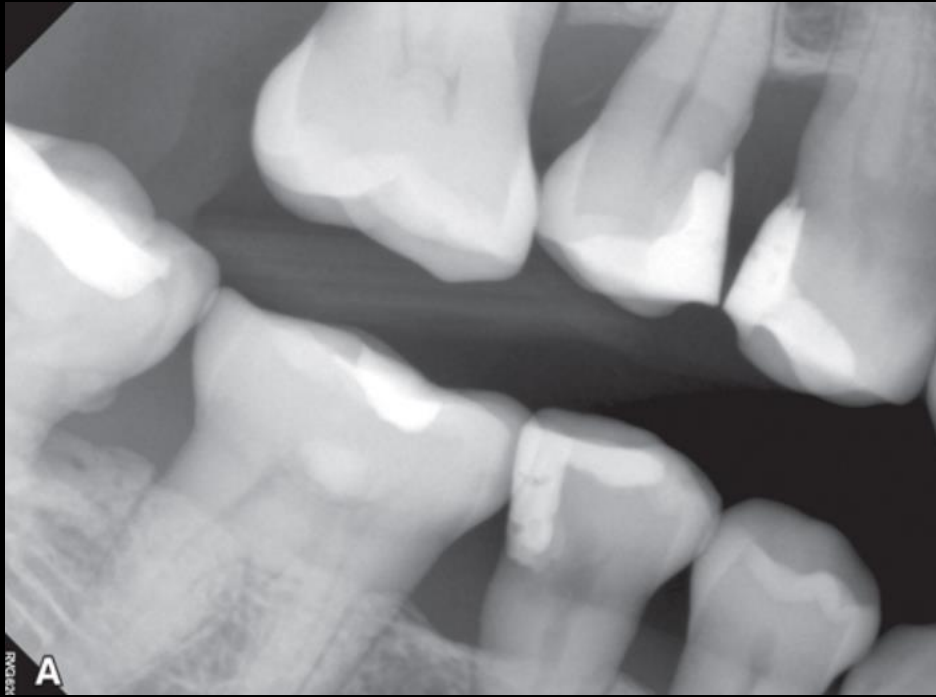
Radiographic Assessment of Periodontal Conditions

Radiographs are especially helpful in the evaluation of the following features:

- Amount of bone present
- Condition of the alveolar crests
- Bone loss in the furcation areas
- Width of the periodontal ligament space
- Local irritating factors that increase the risk of periodontal disease
 - Calculus
 - Poorly contoured or overextended restorations
 - Root length and morphology and crown-to-root ratio
 - Open interproximal contacts, which may be sites for food impaction
- Anatomic considerations
 - Root proximity
 - Position of the maxillary sinus in relation to a periodontal deformity
 - Position of the mental foramen in relation to the alveolar crest

Intraoral Imaging for Periodontal Assessment

- **Highest spatial resolution** → detects subtle changes (PDL space, lamina dura, early bone loss)
- **Bitewing images:**
 - Best for assessing CEJ–alveolar crest distance
 - Use **vertical bitewings** for patients with significant bone loss
- **Periapical images:**
 - Show full tooth length
 - May distort CEJ–crest relationship (especially in maxilla)



Limitations of Intraoral Imaging

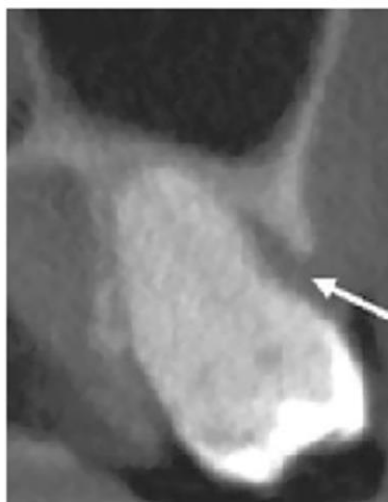
- **2D limitations:** Cannot show buccal/lingual bone defects clearly
- **Superimposition:** May obscure crestal bone
- **Underestimates bone loss**
- **No soft tissue info:** Cannot assess pocket depth
- **CEJ as reference:** Not reliable in supraeruption or attrition

Panoramic Imaging & Extraoral Bitewings

- **Panoramic imaging:**
 - Quick, comfortable, full-jaw overview
 - **Lower resolution, distortion, and superimposition**
 - Not recommended as primary tool for periodontal diagnosis
- **Extraoral bitewings:**
 - Newer option in some panoramic systems
 - **No current evidence** supports their use for periodontal assessment

CBCT in Periodontal Assessment

- **Advantages:**
 - 3D visualization of **osseous structures**
 - No **anatomical superimposition**
 - Better assessment of:
 - **Vertical defects**
 - **Furcations**
 - **Buccal/lingual bone loss**
 - **Root morphology, fenestrations, periapical disease**
- **Limitations:**
 - **Artifacts** from metallic restorations
 - **Higher cost, radiation dose, and time**
 - **Not routinely recommended** for periodontal diagnosis
- **Indications:**
 - **Small-volume, high-resolution CBCT** may be used for **surgical planning** or complex bony lesions



Normal Radiographic Anatomy of the Periodontium

- **Alveolar crest:**
 - Radiopaque cortical bone
 - Located **0.5–2.0 mm apical** to CEJ
- **Posterior teeth:** Crest parallels CEJ line
- **Anterior teeth:** Crest appears **peak-like**
- **PDL space:** Thin, uniform; may widen near cervical root in youth
- **Variations** (e.g., thin crests, radiolucency) may be **normal**



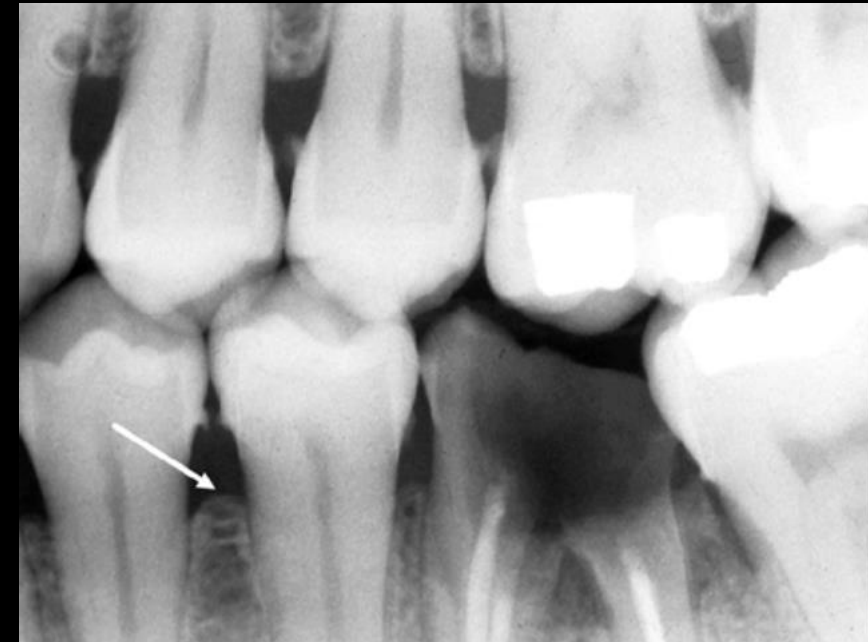
Radiographic Features of Periodontal Disease

- **Gingivitis:** No radiographic bone changes
- **Periodontitis:**
 - **Morphologic changes:** Loss of interproximal bone, cortical plate
 - **Trabecular changes:**
 - **Rarefaction:** Increased radiolucency (bone loss)
 - **Sclerosis:** Increased radiopacity (bone formation)
- **Acute** = bone loss; **Chronic** = sclerosis + loss

Changes in morphology of alveolar processes

Early and Horizontal Bone Loss

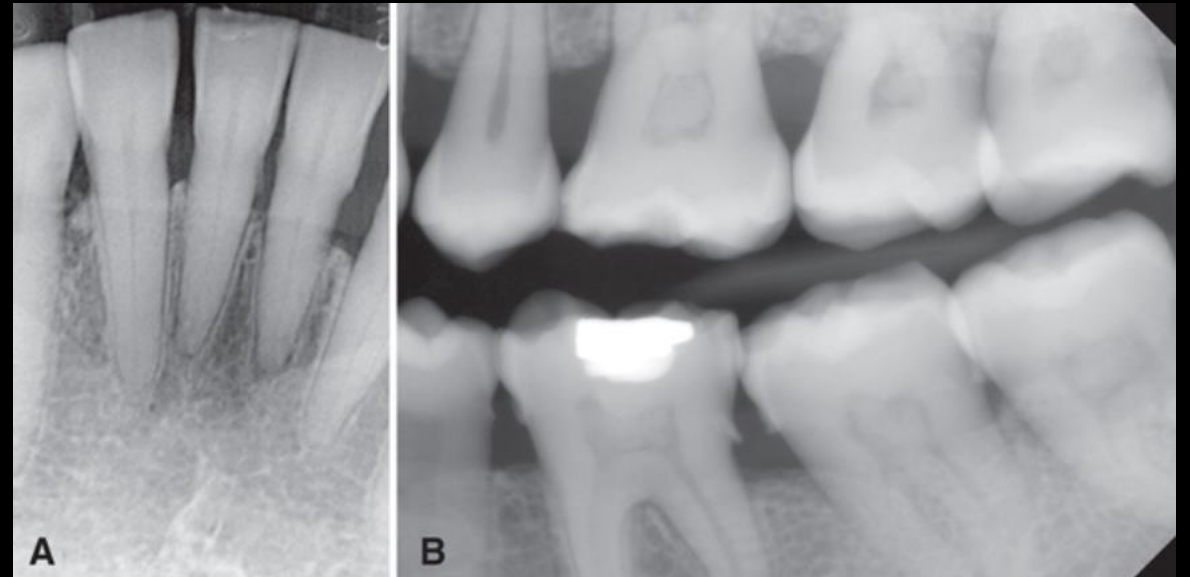
- **Early Signs of Bone Loss:**
 - **Blunting** of anterior alveolar crests
 - **Rounding** of the angle between posterior crest and lamina dura
 - **Diffuse cortical bone edge** (less defined)

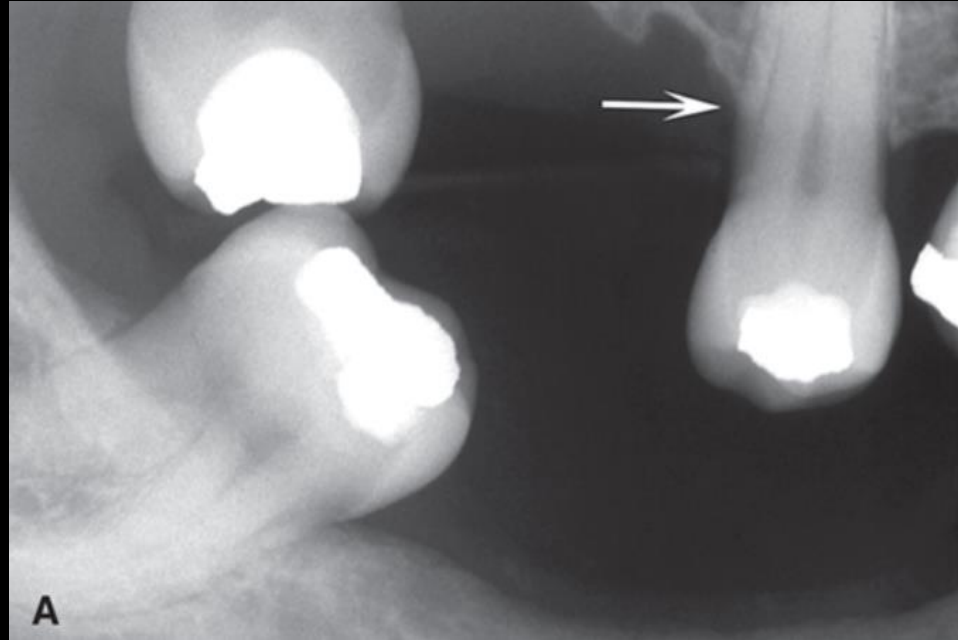
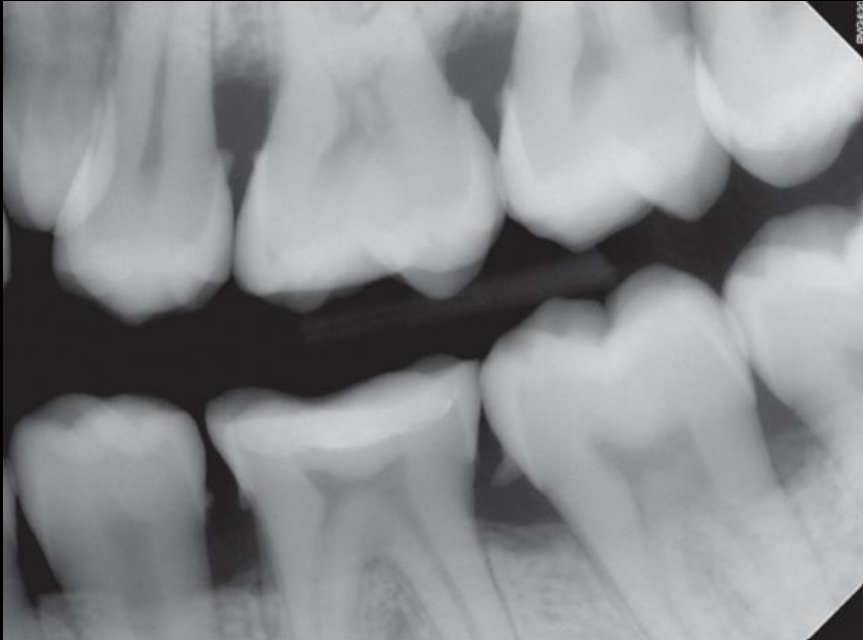


Changes in morphology of alveolar processes

Early and Horizontal Bone Loss

- **Horizontal Bone Loss:**
 - Bone crest remains **parallel to CEJ line**
 - Severity categorized by bone level relative to **root length**:
 - **Mild:** within coronal 15%
 - **Moderate:** 15–33%
 - **Severe:** beyond 33%





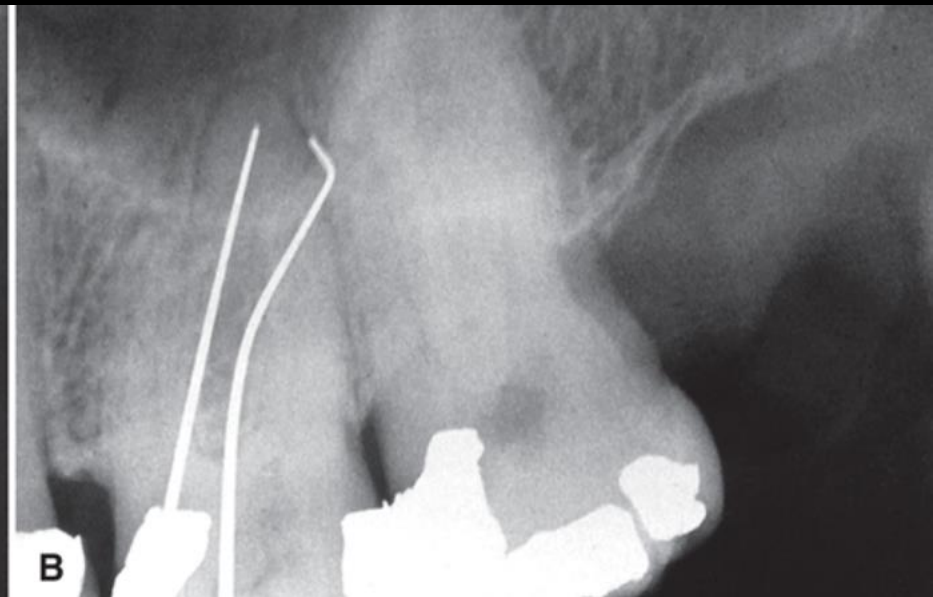
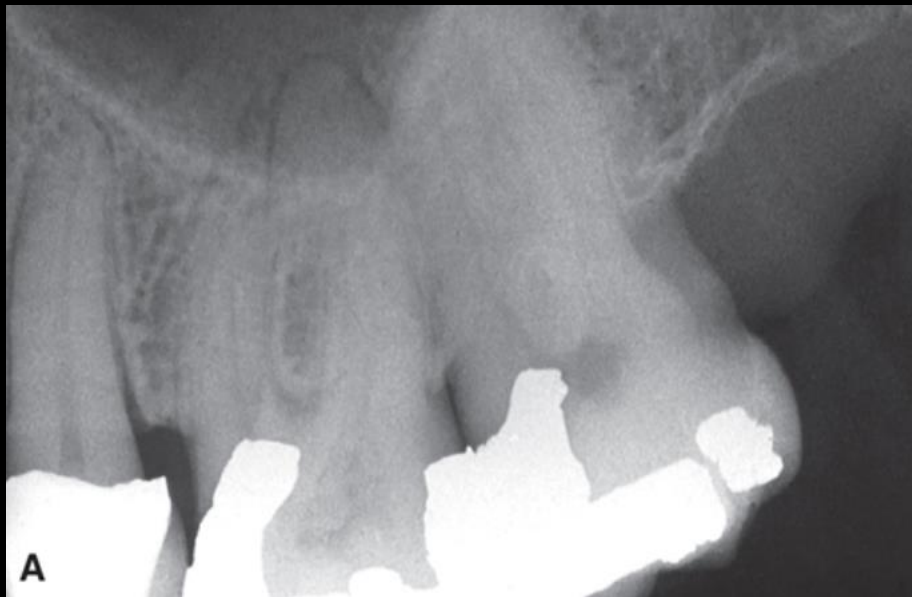
Changes in morphology of alveolar processes

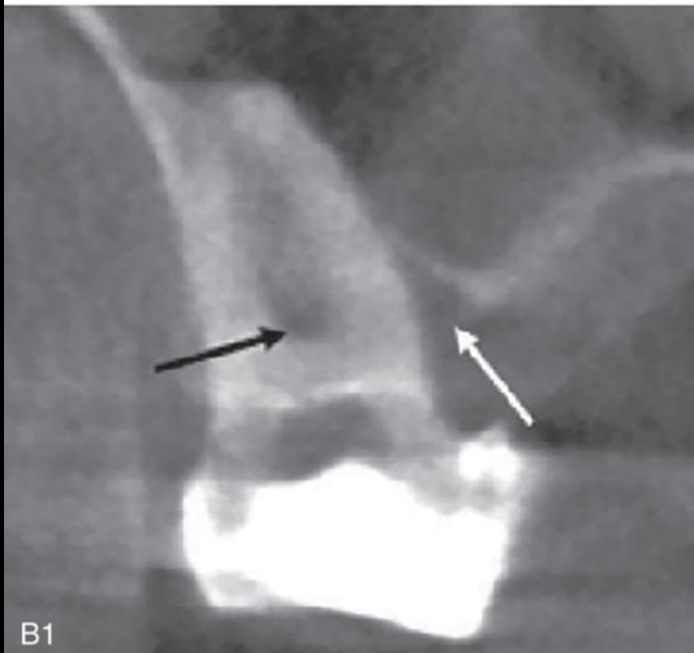
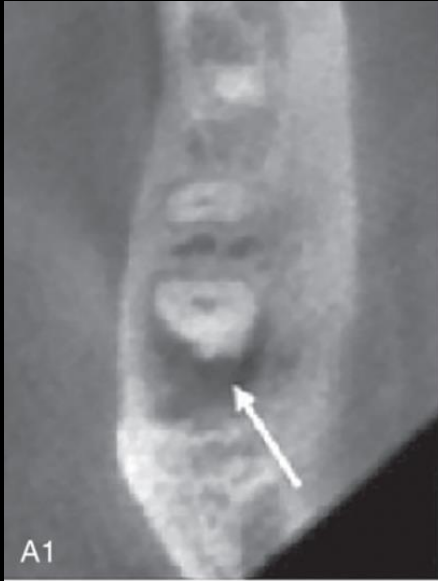
Vertical Bone Defects

- **Vertical (angular) defect:** V-shaped bone loss along root surface
- Appears **oblique** to CEJ line on radiographs
- **Early sign:** Widened PDL at crest
- Classified by number of remaining bony walls:
 - **3-walled:** buccal & lingual plates intact
 - **2-walled:** one plate lost
 - **1-walled:** both plates lost
- **Gutta-percha tracing** or **CBCT** may aid visualization



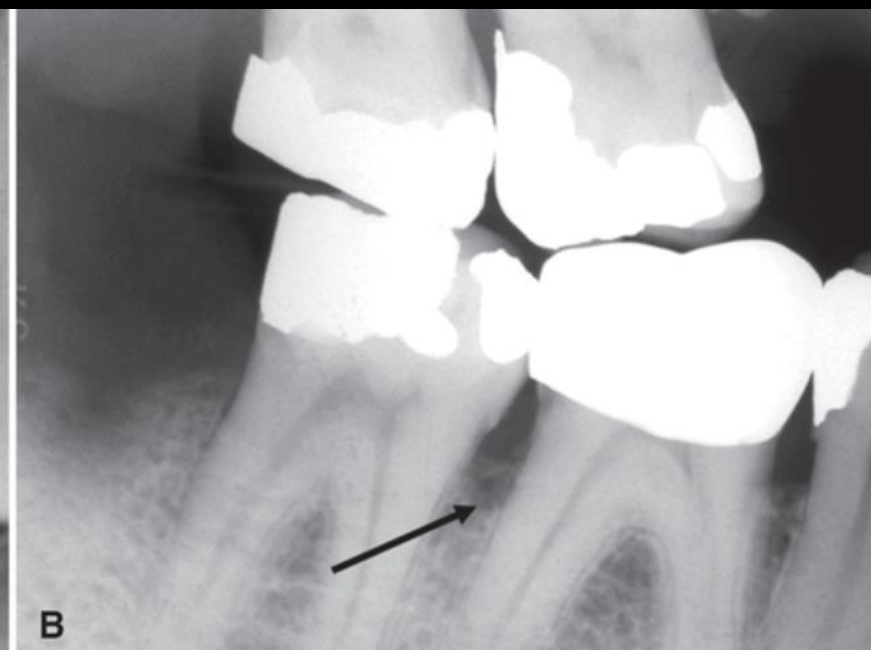
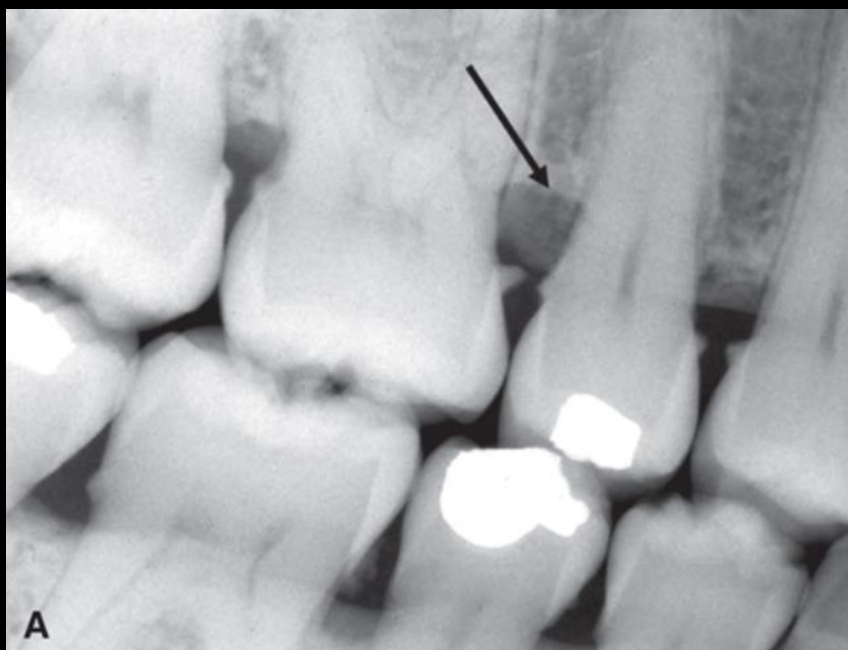


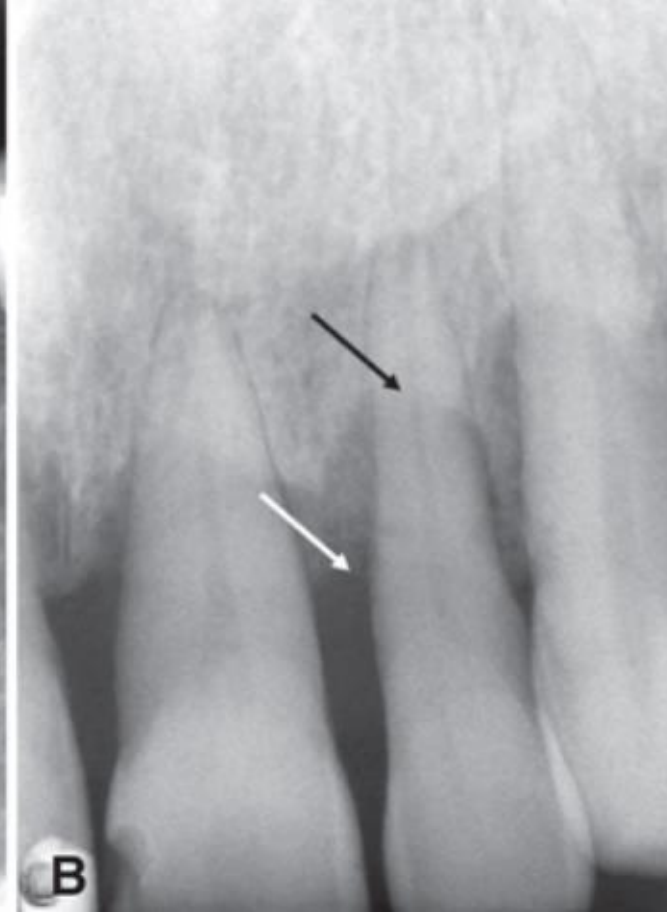




Interdental Craters & Cortical Plate Loss

- **Interdental craters:**
 - 2-walled defects between adjacent teeth
 - Appear as **band-like radiolucency** at crest
 - Common in **posterior jaws**
- **Buccal/Lingual cortical plate loss:**
 - Appears as **semicircular radiolucency** near crest
 - May be hard to detect without interproximal bone loss



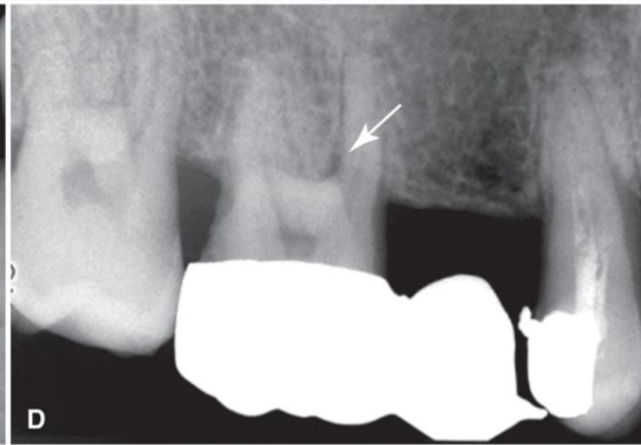
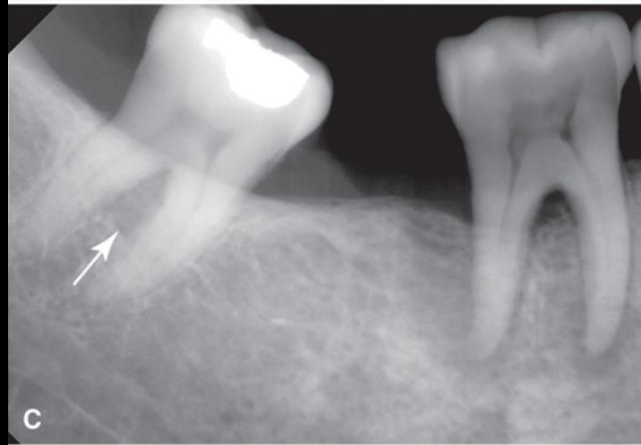
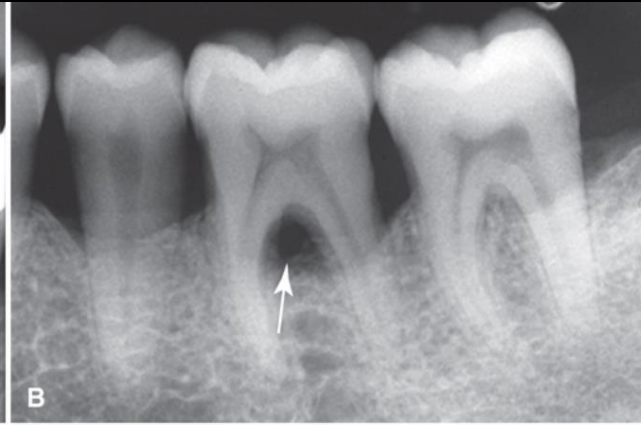
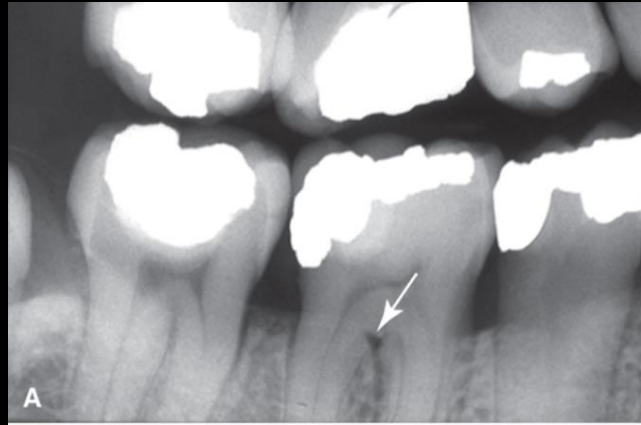


Furcation Involvement in Multirrooted Teeth

- Bone loss may extend into **furcations** of molars
- **Mandibular molars**: clearer furcation visibility
- **Maxillary molars**: palatal root may obscure defect
- Radiographic signs:
 - **Widened PDL** at furcation
 - **Radiolucent triangle** or **inverted J-shape**
- **CBCT** or **surgical inspection** may be needed for full assessment

Advanced Periodontal Bone Loss Patterns

- **Vertical defects:** localized, V-shaped, classified by wall number
- **Craters:** 2-walled, common in posterior interproximal bone
- **Cortical plate loss:** semicircular radiolucency, subtle
- **Furcation defects:** radiolucent signs vary by tooth and root anatomy
- **CBCT:** useful but not routine; reserved for complex cases



Severe periodontal bone loss



Periodontal attachment loss can ONLY be evaluated clinically!!

Bone Density & Trabecular Pattern Changes in Periodontal Disease

- **Radiolucent changes:**
 - Loss of trabecular density
 - Faint or invisible trabeculae (early/acute inflammation)
 - May **remineralize** with successful treatment
- **Radiopaque (sclerotic) changes:**
 - Thickened trabeculae → increased radiopacity
 - May appear **amorphous** in severe cases
 - Sclerosis may extend to **mandibular cortex** or **sinus floor**
- **Other findings:**
 - **Maxillary sinus mucositis** from inflammatory spread
 - Rare **periosteal reactions** in adjacent bone



Classification of Periodontal Diseases

- Classification based on:
 - **Tissue involvement** (gingiva vs. bone)
 - **Staging**: Severity & complexity
 - **Grading**: Rate of progression & risk
 - **Systemic associations**
- **Gingivitis**: Inflammation of gingiva only
- **Periodontitis**: Includes attachment and bone loss
- Gingivitis **precedes** periodontitis, but doesn't always progress

Periodontal Disease Classification

1. Periodontal health, gingival diseases and conditions

- a. Periodontal health and gingival health
- b. Gingivitis: dental biofilm induced
- c. Gingival diseases: nondental biofilm induced

2. Periodontitis

a. Necrotizing periodontal diseases

- i. Necrotizing gingivitis
- ii. Necrotizing periodontitis
- iii. Necrotizing stomatitis

b. Periodontitis

i. Stages

- Stage I: Initial periodontitis
- Stage II: Moderate periodontitis
- Stage III: Severe periodontitis with potential for additional tooth loss
- Stage IV: Severe periodontitis with potential for loss of the dentition

ii. Extent and distribution: localized; generalized; molar–incisor distribution

iii. Grades: Evidence or risk of rapid progression, anticipated treatment response

- Grade A: Slow rate of progression
- Grade B: Moderate rate of progression
- Grade C: Rapid rate of progression

c. Periodontitis as a manifestation of systemic disease

3. Other conditions affecting the periodontium

- a. Systemic diseases or conditions affecting the periodontal support system
- b. Periodontal abscesses and endodontic-periodontic lesions
- c. Mucogingival deformities and conditions
- d. Traumatic occlusal forces
- e. Tooth and prosthesis-related factors

Periodontitis Classification

Periodontitis Stage		I	II	III	IV
Severity	Radiographic bone loss	Coronal third of root (<15%)	Coronal third of root (15%–33%)	Extending to the middle third of root and beyond	Extending to the middle third of root and beyond
Complexity	Local	Maximum probing depth ≤4 mm that is mostly horizontal bone loss	Maximum probing depth ≤5 mm that is mostly horizontal bone loss	Stage II complexity and probing depth ≥6 mm; vertical bone loss ≥3 mm; Class II or III furcation; moderate ridge defect	Stage III complexity and severe ridge defect and <20 remaining teeth (10 opposing pairs)

Subtypes of Periodontitis & Grade Modifiers

- **Subcategories:**
 - **Necrotizing periodontal diseases (NPDs)**
 - **Periodontitis**
 - **Systemic disease–associated periodontitis**
- **Grade modifiers:**
 - **Diabetes mellitus:** Bidirectional relationship
 - **Smoking:** Alters microflora, impairs immunity

Necrotizing & Systemic Periodontal Diseases

- **Necrotizing Periodontal Diseases (NPDs):**
 - Severe pain, fever, ulceration, necrosis
 - Risk factors: HIV, malnutrition, stress, smoking
- **Systemic disease–associated periodontitis:**
 - Linked to conditions like:
 - Leukemia, neutropenia
 - Down syndrome, Papillon–Lefèvre syndrome
 - Often due to **immune dysfunction** or **connective tissue defects**

Other conditions affecting the periodontium

Periodontal Abscess

- **Radiographic features:**

- May be **absent** in acute phase
- **Chronic abscess:** radiolucent area superimposed over root
- **Loss of lamina dura** at root surface
- May show **bone bridge** over lesion, separating it from alveolar crest

- **Healing:**

- Bone may **regenerate** after treatment



Endodontic-Periodontic Lesions

- May originate from:
 - **Periodontal lesion** extending to apex → secondary pulpitis
 - **Periapical lesion** extending coronally → retrograde periodontitis
- Radiographic appearance:
 - **Deep angular defect** extending to apex
 - May show **periapical rarefying osteitis**
 - **Funnel-shaped** bone loss
- Treatment: Requires **both endodontic and periodontal therapy**



Occlusal Trauma & Tooth Mobility

- **Occlusal trauma:**
 - Caused by excessive or misdirected occlusal forces
 - Imaging signs: **Widened PDL, thickened lamina dura, bone loss, hypercementosis**
 - Does **not cause periodontitis**, but may **accelerate bone loss** if disease is present
- **Tooth mobility:**
 - Seen as **hourglass-shaped socket** (single-rooted teeth)
 - **Widened PDL** at apex and furcation (multi-rooted teeth)
 - Lamina dura may appear **broad and dense**

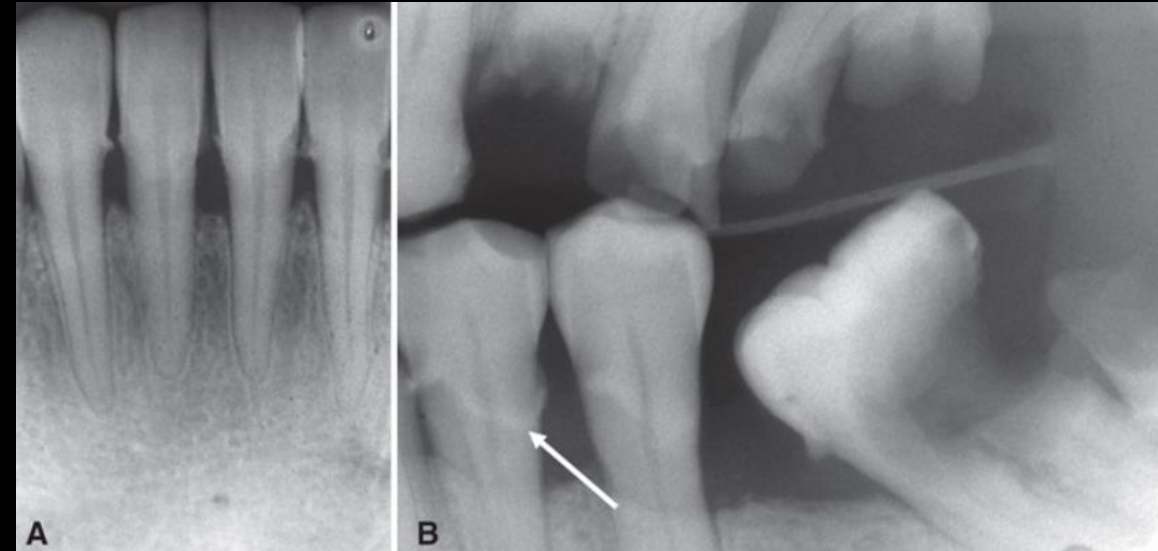
Open Contacts & Malalignment

- **Open contacts:**
 - Increase risk of **food impaction**
 - Can lead to **localized inflammation and bone loss**
- **Malalignment:**
 - Includes **tipped teeth, uneven marginal ridges**
 - Doesn't cause disease directly but **hinders hygiene**
 - Creates environment conducive to **plaque accumulation**



Local Irritating Factors

- **Calculus:** Promotes plaque retention, common near mandibular incisors
- **Defective restorations:** Overhangs, poor contours trap plaque
- **Enamel pearls & cervical enamel projections:** Common in furcations, disrupt attachment





Evaluation of Therapy

- **Post-Treatment Imaging:**
 - Signs of healing: **Re-cortication, sclerosis, remineralization**
 - **Beam angulation & exposure** affect image interpretation
 - **No visible change** ≠ no healing → clinical exam is essential



Differential Diagnosis of Bone Loss

- **Periodontal bone loss:**
 - Starts at **alveolar crest**, progresses **apically**
 - Appears **wider coronally**, tapering apically
- **Red flags for non-periodontal disease:**
 - **Isolated tooth mobility**
 - **Irregular bone loss** not following periodontal patterns
 - **Ill-defined borders, no sclerosis**
- **Consider:**
 - **Squamous cell carcinoma:** Irregular PDL widening, lamina dura destruction
 - **Langerhans cell histiocytosis:** “Floating teeth,” mid-root epicenter





